

CONTACT

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Kerala, India
LinkedIn
GitHub

TECHNICAL SKILLS

- Languages:**
Go, Python, Java, JavaScript, C
- Backend:**
Django, Spring Boot, Node.js, Express.js, REST APIs, WebSockets
- Databases:**
PostgreSQL, MySQL, MongoDB
- DevOps & Cloud:**
Docker, Git, CI/CD, AWS EC2, Microservices
- Systems & Architecture:**
Concurrency, WebSockets (STOMP)

EDUCATION

B.Tech in Computer Science
Christ College of Engineering,
Irinjalakuda
2024 – 2028
CGPA: 9.0/10

HONORS & AWARDS

- SIH'25 Finalist - National hackathon
- 1st Prize - Idea Pitching Challenge
- 2nd Prize - Synovate 2025 Ideathon
- 2nd Prize - Sustainable Dev. Comp.

LEADERSHIP

- Doc Lead, IEEE WIE CCE**
Managed end-to-end technical documentation and reporting workflows.
- Event Coord., FOSS Club**
Directed logistics, judging, and faculty coordination for multiple tech events.
- Publicity, Beach Hack S7**
Drove engagement and participant outreach for a national-level hackathon.

ALIN ALEX
Backend & Software Engineering Enthusiast

PROFESSIONAL SUMMARY

Backend and full-stack focused Computer Science student with hands-on experience in real-time systems, satellite-image analytics, and concurrent processing. Built MineGuard for illegal mining detection, developed FoodHub for real-time restaurant synchronization, and engineered scalable backend architectures. SIH'25 finalist seeking backend or software engineering opportunities.

WORK EXPERIENCE

- Whitematrix | Software Development Intern** 2026
- Spearheaded technical documentation and architectural blueprints for an AI-powered SQL engine (NLIDB), accelerating team onboarding and reducing development bottlenecks by 30%.
 - Defined the backend pipeline across Librarian, Specialist, and Executor layers, implementing dynamic schema injection to achieve 40% faster deterministic SQL generation.
 - Documented secure, read-only database execution (SQLAlchemy/PostgreSQL) and structured Docker/AWS deployment plans, reducing system hallucination risks by 85%.
- Ziuke Infotech | Full Stack Developer Intern** 2025
- Developed “Cart Shopee,” a robust e-commerce application handling 500+ products and active user accounts, leveraging Django ORM to improve database query efficiency by 30%.
 - Engineered real-time cart synchronization by integrating Django backend logic with dynamic frontend updates, reducing order processing errors by 95% and ensuring 100% transaction accuracy.

TECHNICAL PROJECTS

- MineGuard | Satellite-Based Illegal Mining Detection System**
- Engineered a satellite analysis system utilizing Sentinel-1/2 data to detect illegal open-cast mining, reducing false positives by 90%.
 - Designed a “Triple-Lock Verification Model” combining NDVI/NDBI spectral analysis, DEM topographic validation for excavation depth, and GeoJSON geo-fencing.
 - Built a 3D forensic visualization dashboard using Plotly to map DEM elevation data into actionable structural pit evidence.
- Concurrent Job Processing System | Go Architecture**
- Built an asynchronous backend job processing architecture utilizing Goroutines, channels, and worker pools for highly concurrent task execution.
 - Implemented fault-tolerant features including advanced retry mechanisms with exponential backoff and jitter.
 - Used context cancellation for graceful shutdown and precise lifecycle control.
- FoodHub | Real-Time Restaurant Management**
- Developed a real-time order state management system utilizing Spring WebSockets (STOMP) for millisecond-latency synchronization across Kitchen and Server dashboards.
 - Designed a robust relational database schema with Join Tables to track historical pricing anomalies and manage Many-to-Many relationships.
 - Implemented role-based message routing to securely filter websocket events based on strict user permissions.
- E-Ballot | Secure Voting Platform**
- Architected a secure voting platform employing a “Defense in Depth” strategy with bcrypt hashing and JWT-based session management via httpOnly cookies.
 - Enforced structural database integrity through Unique Indexing to strictly prevent double-voting.